

Week 7 – Introduction to Statistics for Data Science

Statistical Business Analysis

Project Overview

This project focuses on applying statistical methods to business data using Python. The analysis includes descriptive statistics, hypothesis testing, correlation analysis, confidence intervals, and regression analysis.

Objectives

- Calculate descriptive statistics.
- Analyze business data distributions.
- Perform hypothesis testing.
- Analyze correlation and covariance.
- Calculate confidence intervals.
- Perform regression analysis.
- Generate useful business insights.

Datasets

Two datasets were used:

- `sales\_data.csv` – Sales transaction data.
- `customer\_churn.csv` – Customer churn and subscription information.

The sales dataset contains 100 records and the customer churn dataset contains 500 records.

Technologies Used

- Python
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Data Preparation

The datasets were loaded using Pandas and checked for:

- Missing values
- Duplicate records

- Data types
- Numerical and categorical columns

The data was cleaned and prepared before performing statistical analysis.

Descriptive Statistics

Descriptive statistics were calculated for important numerical variables including:

- Mean
- Median
- Mode
- Standard deviation
- Minimum
- Maximum
- Quartiles

For the customer dataset:

- Average Tenure: 36.53
- Average Monthly Charges: 113.64
- Average Total Charges: 4237.88

Hypothesis Testing

Three statistical tests were performed.

1. Independent T-Test

The average Monthly Charges of customers who churned and customers who did not churn were compared.

- H0: There is no significant difference in average monthly charges.
- H1: There is a significant difference.
- p-value: 0.0101
- Significance level: 0.05

Result: The null hypothesis was rejected.

This indicates a statistically significant difference in monthly charges between the two customer groups.

2. Chi-Square Test

Customer Contract type and Churn were analyzed.

- H0: Contract type and churn are independent.
- H1: Contract type and churn are associated.

- p-value: < 0.001
- Significance level: 0.05

Result: The null hypothesis was rejected.

This indicates a significant relationship between contract type and customer churn.

3. Pearson Correlation Test

Tenure and Total Charges were analyzed using Pearson correlation.

- Correlation coefficient: -0.0057
- p-value: 0.8992
- Significance level: 0.05

Result: The correlation was not statistically significant in this dataset.

Correlation Analysis

Pearson correlation and covariance were used to understand relationships between numerical variables.

A correlation heatmap was created to visually represent relationships between variables such as:

- Tenure
- Monthly Charges
- Total Charges

Confidence Interval

A 95% confidence interval was calculated for appropriate business metrics to estimate the range in which the population parameter is expected to fall.

The margin of error was also calculated and interpreted as part of the statistical analysis.

Regression Analysis

Linear regression was used to study the relationship between relevant numerical variables.

The regression analysis includes:

- Regression coefficient
- Intercept
- R-squared
- Statistical significance

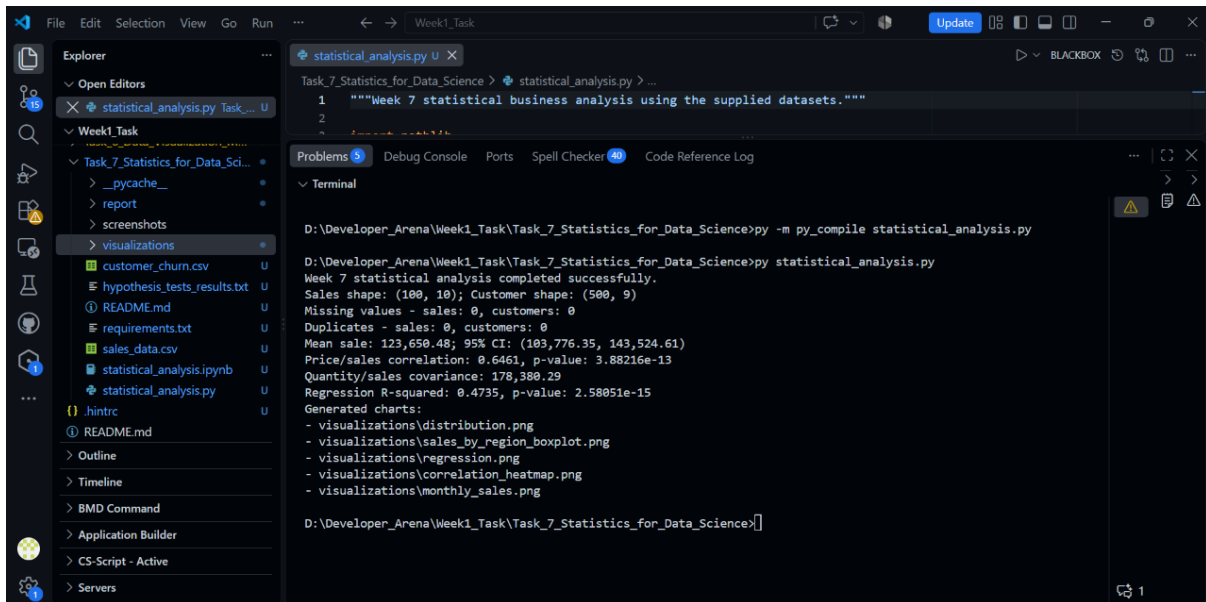
The results were used to understand how one business variable relates to another.

Visualizations

The project includes statistical visualizations such as:

1. Distribution Histogram
2. Box Plot
3. Correlation Heatmap
4. Regression Scatter Plot

These visualizations help explain the statistical results clearly.



Business Insights

- Customers with higher average monthly charges showed a statistically significant difference in churn behavior.
- Contract type has a statistically significant relationship with churn.
- The analyzed relationship between tenure and total charges was not statistically significant.
- Statistical analysis can help identify customer groups that may require better retention strategies.

Conclusion

This project demonstrates how statistical methods can be applied to real business datasets.

Descriptive statistics, hypothesis testing, correlation, confidence intervals, and regression provide useful methods for understanding customer behavior and business performance.

The analysis provides a statistical foundation for making data-driven business decisions.